1 Introduction

Breast cancer is one of the most prevalent and life-threatening diseases affecting women worldwide. According to the World Health Organization/International Agency for Research on Cancer (2022), breast cancer is the most common cancer worldwide and accounts for one in four cancer cases among women in 2020 with a 30% mortality rate. Closer to our academic institution, it is the most common diagnosed cancer in women in Australia, making up 28% of cancer diagnosis in females (Australian Institute of Health and Welfare, 2023). Early detection and accurate diagnosis of breast cancer are critical for improving survival rates and treatment outcomes, as seen in a 99% 5-year relative survival rate mentioned by the American Cancer Society (2023). The diagnosis of breast cancer typically starts with a medical history review and physical examination, looking out for abnormalities in the breasts and lymph nodes in the armpit and above the collarbone. The doctor will then request for imaging tests if further investigation is required. If a suspicious mass is found in the test, a biopsy will have to be performed to ascertain the pathology, whether it is benign or malignant. As such, medical imaging such as mammography, ultrasound and Magnetic Resonance Imaging (MRI), has played a pivotal role in the early diagnosis of breast cancer, providing valuable insights about the tissue hidden under the skin. That said, mammography has often been the gold standard for it (Barba et al., 2021).

The motivation behind this study lies in the need for more effective and reliable breast cancer screening methods. While mammography remains a valuable tool, there is room for improvement in terms of accuracy, consistency, and speed. By leveraging the power of CNNs, we aim to develop a computer-based system that can assist healthcare professionals in the early detection of breast cancer. Such a system has the potential to reduce false negatives and positives, leading to better patient outcomes and resource utilization.

Additionally, to address the challenges of privacy of medical data and data centralization, we propose the integration of FL into our approach. FL allows for model training across multiple decentralized data sources (e.g. different healthcare institutions) without the need to share sensitive patient data. This approach ensures patient privacy while still benefiting from a bigger, collective dataset.

The primary objective of this research is to investigate the integration of AI in breast cancer diagnosis with mammograms. Specifically, we aim to:

- Study publicly available mammogram datasets, indicating characteristics of each dataset and highlighting class imbalance if any.
- Evaluate the performance of current state-of-the-art (SOTA) models in this domain of mammography classification.
- Investigate the potential for integrating the CNN-based system with FL to enhance privacy-preserving model training and to accommodate decentralized data sources, while achieving comparable or superior performance compared to centralized training.

The rest of the paper is structured in the following way to enhance reader accessibility and foster a coherent comprehension of the topic - Section 2 will provide some background information necessary in understanding the disease and the diagnosis process. Section 3 highlights the publicly available datasets that allows researchers to utilize for training machine learning models. Section 4 will discuss related works pertaining to this task, and Section 5 talks about the methods used in conducting the experiments. Following that, Section 6 shows the technical implementation done during this study, classifying mammograms benign or malignant. This was also executed in an FL setting to explore the possibility of extending this domain to a decentralized environment. Lastly, Section 7 summarizes all that have been done in this study, highlighting future works and possibly directions of where mammogram classification can move towards.

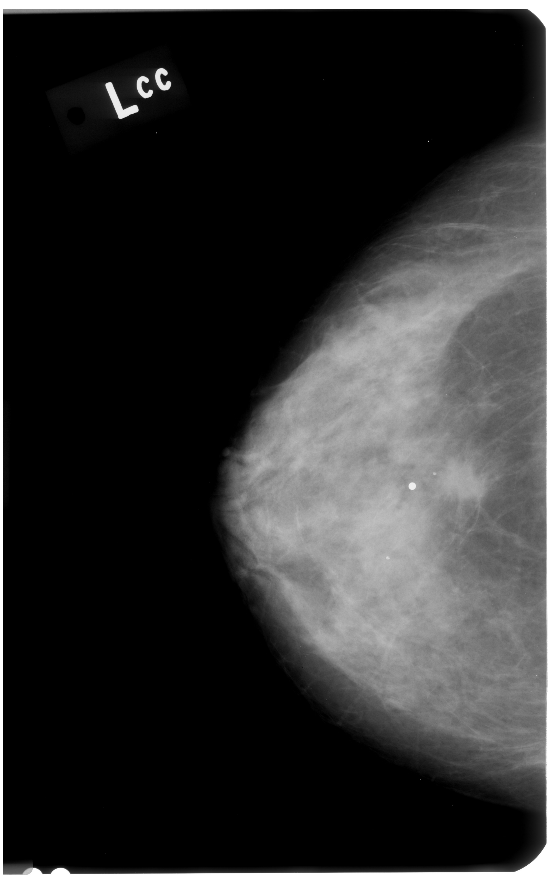
2 Background

2.1 Mammograms

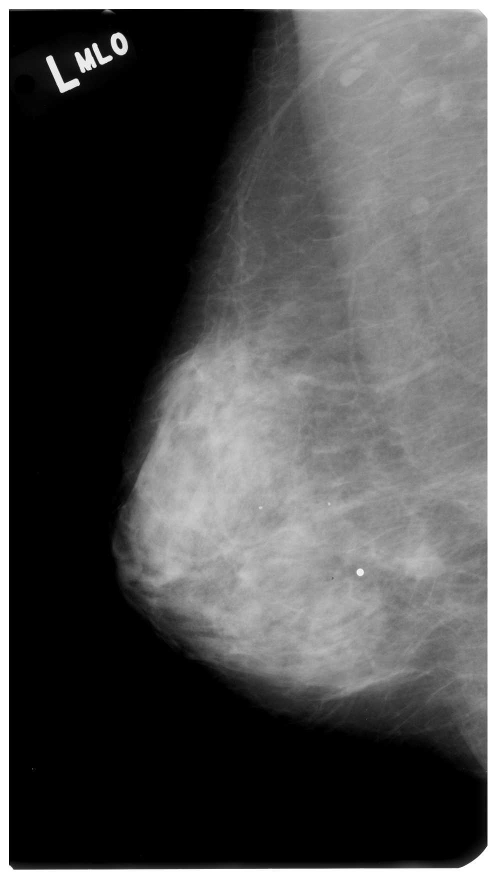
Mammography, a low-dose X-ray imaging technique, has been the primary screening tool for breast cancer for many years. It allows radiologists to visualize breast tissue and identify abnormalities, such as tumors or suspicious masses. However, the process of interpreting mammograms is highly dependent on the expertise of radiologists, and the task can be time-consuming and subject to human error. Furthermore, the noise, low contrast and artifacts in mammograms can lead to differing diagnosis (Elmore et al., 2009). This hinders early detection and results in late treatment of the illness due to false negatives, or unnecessary patient anxiety due to false positives. The demand for accurate and efficient breast cancer screening methods has thus led to the exploration of computer-aided diagnosis (CAD) systems (Elter & Horsch, 2009).

Film mammography was regarded as the gold standard ever since its invention in 1970s (Miller, 2001). However, it was less effective for women with denser breast tissue (Checka et al., 2012). The development of mammography rode the digitalization of society, which led to digital mammography. Digital mammography showed comparable diagnostic accuracy and superiority in various aspects such as imaging processing, detection, and resolution, particularly benefiting women with dense breasts and younger individuals (Faridah, 2008). It has also been found that going digital gave rise to higher detection accuracy (Turco et al., 2007). Digital mammography gave the ability to manipulate and enhance images, thus offering promising techniques such as contrast-enhanced mammography and tomosynthesis. The latter, given its 3D imaging, has found to give better performances compared to digital mammography in recent deep learning studies (Li et al., 2019).

In mammography, there are two views for each breast, namely the bilateral craniocaudal (CC) view and mediolateral oblique (MLO) view. The CC view refers to a top view of the breast, while MLO refers to a side view of the breast. As seen in Figure 1b, the pectoral muscle is usually more apparent in the MLO view compared to the CC view.



(a) CC View



(b) MLO View

Figure 1: Comparison between CC and MLO View

Table 1: BI-RADS Assessment Categories

BI-RADS	Risk of Malignancy	Description
0	Further imaging required	Incomplete evaluation, additional mammographic views, spot compression, magnification, or ultrasound needed
1	0%	Negative examination, no masses, calcifications, or architectural distortion
2	0%	Consistent with benign findings such as secretory calcifications, cysts, etc.
3	$<$ 2%	Probably benign with strict criteria for non-palpable, circumscribed mass or focal asymmetry
4a	2-10%	Low probability of malignancy
4b	10-50%	Intermediate probability of malignancy
4c	50-95%	High probability of malignancy
5	$>$ 95%	Highly suggestive of malignancy
6	100%	Pathology proven malignancy

Another property of mammograms is the BI-RADS (Breast Imaging-Reporting and Data System) value which is indicated by radiologists to deem whether further assessments are necessary, ranging from 0 to 6 (Magny et al., 2023). BI-RADS 1 would indicate approximately zero risk of malignancy, while BI-RADS 5 would indicate > 95% risk of malignancy, as seen in Table 1.

Breast cancer can exist as a mass or a calcification (which is a build-up of calcium in a tissue). Figure 2 shows the visual difference between the two - calcifications appear as tiny white dots or spots on imaging, whereas masses appear as distinct, often larger, rounded areas.

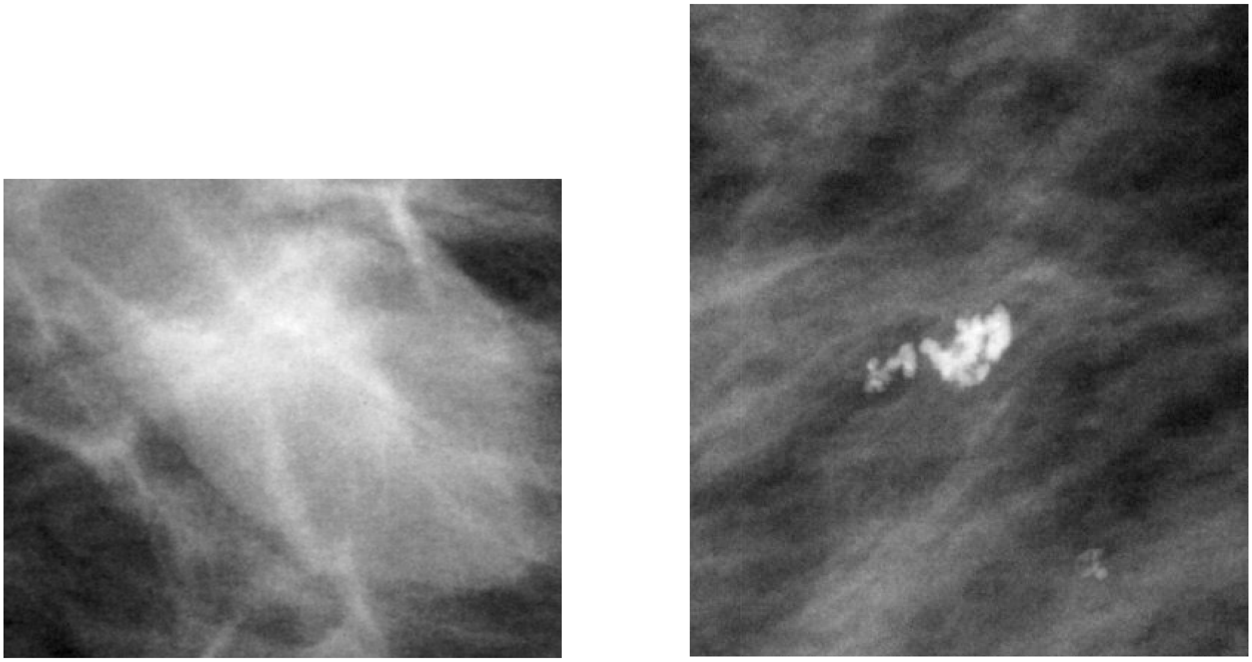
From prior research, it has been determined that ethnicity contribute to the likelihood of developing breast cancer (Hirko et al., 2022). Ethnic minority women were found to have less favourable tumour characteristics, possibly due to certain inherited genetic mutations that would increase risk of breast cancer, or lifestyle factors (Gathani et al., 2021). Barriers to early detection and healthcare treatments and lack of research for these minority groups were also mentioned to be causes for delayed diagnosis for women in these groups (Yedjou et al., 2019).

2.2 Deep Learning

In recent years, deep learning techniques, especially Convolutional Neural Networks (CNNs), have shown remarkable success in various image analysis tasks, including object recognition, image segmentation, and medical image classification (Khan et al., 2020). CNNs have demonstrated their potential to assist radiologists in interpreting medical images, including brain computed tomography, achieving equivalent diagnostic performances to that of radiologists (Jørgensen et al., 2022). CNNs require large amounts of data to be highly effective in making accurate diagnosis, which pose a challenge when training them on medical data due to the scarcity of publicly available labelled data. Strategies to allow CNNs to perform on small medical datasets include transfer learning and data augmentation (Yamashita et al., 2018).

Transfer learning is a pivotal paradigm in deep learning that leverages knowledge learned from one task to improve performance on a related task (Pan & Yang, 2010). It capitalizes on the idea that models pre-trained on large datasets such as ImageNet with 1.2 million images, can acquire valuable generalizations and feature representations that are transferrable across domains or tasks (Yosinski et al., 2014). This approach not only accelerates model development but also enables better performance on small datasets (Donahue et al., 2013).

Data augmentation is another technique used in deep learning that involves applying a variety of



(a) Mass Lesion (b) Calcification Lesion

Figure 2: Comparison between Mass and Calcification

transformations to the dataset, such as rotation, cropping and flipping. These transformations effectively increase the size and diversity of the training data, which is valuable for improving model generalization and robust- ness (Shorten & Khoshgoftaar, 2019). Data augmentation has been widely adopted in computer vision tasks with small training datasets (Mumuni & Mumuni, 2022), where it helps prevent overfitting and enhances the model's ability to handle variations in real-world data.

Federated Learning (FL) was first introduced in 2016 by Google researchers (Konečný et al., 2016), proposing to bring one centralized model to multiple decentralized devices, train on the data on those devices, and then returning the trained parameters back to the central server (McMahan et al., 2023a). This allows the model to aggregate knowledge learnt from a large number of data sources without having to extract data from these devices, promoting control over private data while allowing for collaborative machine learning.

3 Datasets

Several publicly available mammography datasets have been studied to identify useful characteristics for deep learning models to leverage on. These include the Curated Breast Imaging Subset of Digital Database for Screening Mammography (CBIS-DDSM) (Lee et al., 2017), The Chinese Mammography Database (CMMD) (Cai et al., 2023), VinDr-Mammo (Nguyen et al., 2023), Radiological Society of North America (RSNA) (Carr et al., 2022) and Breast Cancer Digital Repository (BCDR) (Guevara Lopez et al., 2012) and Mammographic Image Analysis Society (MIAS) (Suckling et al., 2015).

3.1 CBIS-DDSM

The CBIS-DDSM dataset was employed in this study, offering standardized digitized mammography images in Digital Imaging and Communications in Medicine (DICOM) format. It was curated from several sources including Massachusetts General Hospital, Wake Forest University School of Medicine, Sacred Heart Hospital, and Washington University of St Louis School of Medicine. The dataset contains both CC and MLO views and pixel-level region-of-interest (ROI) labels with annotations. It also classifies them based on their pathology, as well as distinguishes calcifications from masses. It also has the scan's BI-RADS value and breast density specific to each image.

One distinct characteristic of this dataset is its mammogram scan type being of film type. When loading images from this dataset, artifacts, labels and white borders were found present in majority of the images. These made processing of the images difficult, and GradCAM was often found to be distracted by their presence, having a strong heatmap near these areas as seen in Figure 3. Preprocessing the CBIS-DDSM dataset is therefore mandatory when trying to achieve the best performance.

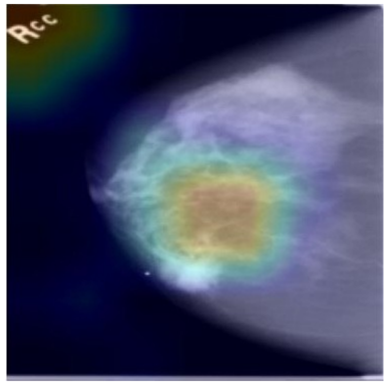


Figure 3: ”RCC” letters affecting the model’s judgement

The dataset is split into four categories: calcification_train, calcification_test, mass_train and mass_test. This is essential given that masses and calcifications are very different visually, allowing research to focus on either forms if necessary. However, scanners used in this dataset was not provided, possibly due to it being collected from four different institutions mentioned earlier.

It is found on the TCIA (The Cancer Imaging Archive) website, together with instructions on how to download their dataset using the NBIA Data Retriever and the manifest file. Alternatively, TCIA has its own REST API (Representational State Transfer Application Programming Interface), that can be used in a Python automation script (Archive, 2023).

3.2 CMMD

The CMMD dataset comprises of 3,712 mammographies from 1,775 patients, divided into two datasets: CMMD1 and CMMD2. CMMD1 consists of 1,026 cases (2,214 mammographies) with biopsy-confirmed benign or malignant tumors, primarily aimed at facilitating the development of Computer-Aided Diagnosis (CADx) and Computer-Aided Detection (CADe) systems. On the other hand, CMMD2 involved 749 patients (1,498 mammographies) with known molecular subtypes, intended for exploring the relationship between image features of invasive carcinoma and molecular subtypes, benefiting from more extensive immunohistochemical markers than CMMD1. Both datasets encompass mammography images and clinical data, such as patient age and tumor diagnosis. It also mentioned the type of lesion that the patient has, between calcifications and mass. The images were digitized at a resolution of 2294 x 1914 pixels and stored in the DICOM format.

The attached metadata file provided pathology information on 1,872 cases. This made the dataset analysis confusing, as it was stated in the paper that 3,712 studies were done on 1,775 patients. It turns out that while the dataset only included one side’s MLO and CC view for most patients (two scans per patient), 97 patients had both left and right sides scanned. This means that this group of 97 has four scans under the same patient ID. Some examples include patients D1-0121, D1-0767, D1-0884, where they have four scans in its folder, but have separate rows in the metadata file for the left and right side.

According to the paper, there are 5,202 images in the whole dataset. However, only 1,872 cases have their confirmed biopsy indicated in the metadata. It was unclear if the remaining 3,330 is assumed to be normal. There is also a lack of ROIs and annotation which may hinder its usage for image segmentation tasks. Furthermore, there is no distinction on which view (CC/MLO) or image had the lesion. This makes it difficult for researchers without clinical knowledge to identify whether the image is being classified correctly or not. In our study, we only used the labelled images as we were not medically trained to determine whether the unlabelled ones are benign or malignant. It is stated by the authors that the type of scanner used for the mammograms was using the GE Senographe DS mammography system.

Similar to CBIS-DDSM, since it is hosted on TCIA, it can be downloaded via the website using their

NBIA Data Retriever or the TCIA API mentioned previously.

3.3 VinDr

The VinDr-Mammo dataset was collected from a pool of mammograms from Hanoi Medical University Hospital (HMUH). It contains 5,000 four-view exams, totaling up to 20,000 images. These images were independently double-read and labelled by experienced mammographers. Each image has its corresponding image view, breast density levels and findings related to abnormalities. This dataset consists of full-field digital mammograms in DICOM format and includes the scanners used for each mammogram.

While the dataset did include ROI bounding boxes, the coordinates of these boxes were only included for BI-RADS 3, 4 or 5. The included metadata did not have explicit diagnosis of each scan between benign and malignant. As such, we used the BI-RADS labels to determine it. As mentioned from its website, BI-RADS 1 and 2 are considered benign, while BI-RADS 3, 4 and 5 are considered malignant in this dataset. It was also not mentioned specifically if a lesion was a mass or a calcification. Similar to CBIS-DDSM, it has its own training and testing split assigned by the authors. Table 2 illustrates the number of scans performed by each designated scanner in this dataset.

Mammography Scanner Name	# Scans
SIEMENS Mammomat Inspiration	15244
Planmed Nuance	3796
IMS GIOTTO CLASS	628
IMS GIOTTO IMAGE 3DL	332

Table 2: Number of scans grouped by scanners in VinDr dataset.

The dataset provided instructions on how to obtain its dataset on its website, either by downloading a ZIP file or the provided CLI command using wget.

3.4 RSNA

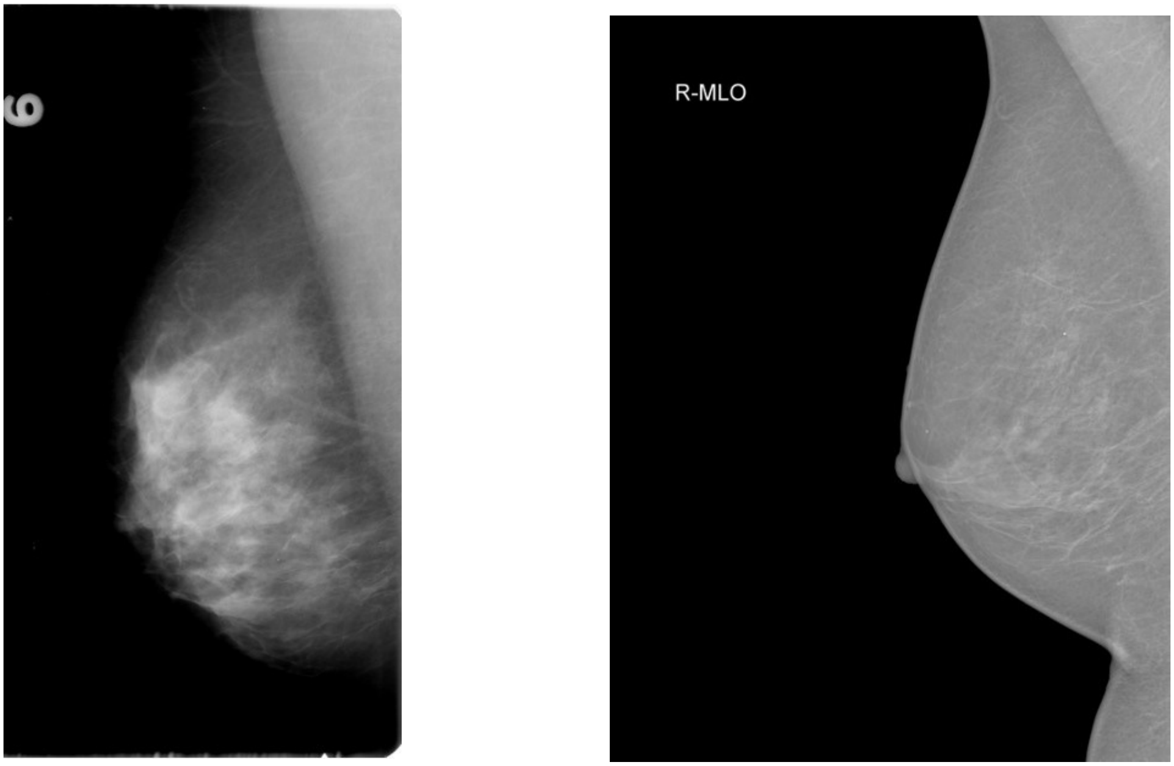
The RSNA Dataset was featured prominently in the 2023 RSNA Screening Mammography Breast Cancer Detection AI Challenge, one of the research competitions organized by RSNA. The dataset was a collaborative effort, with contributions from mammography screening programs in both Australia and the United States. It has a total of 54,700 full-field digital mammographic DICOM files, with 1,158 being malignant and 53,548 being benign. Each patient has at least 4 scans, up to 14 scans in a single folder. However, the metadata that comes with the dataset is thorough and has diagnosis for each of these scans. This is helpful since not all datasets have diagnosis labelling for all 4 scans, making this dataset superior over other datasets with partially labelled data. However, this dataset did not differentiate whether malignant cases were masses or calcifications.

The dataset has a train and test split, but was meant for the Kaggle competition. Since the test folder only had one image, it was decided that only the training folder would be used as the overall dataset. As such, a new training, validation and testing dataset was generated from the original training folder. Also, it does mention a machine_id for every entry, but only the numerical identification was given. There was no glossary that allows readers to identify the actual scanners used.

Kaggle provides the command to use its API to download this dataset.

3.5 Other Comments

It is crucial to take note of the scan types between the studied datasets. All except CBIS-DDSM uses digital mammography, while CBIS-DDSM uses film mammography. The power of digital mammography allows for clearer edges and images to be of higher quality compared to film mammography, as depicted in Figure 4. Properties of each dataset are summarised in Table 4.



(a) Film (b) Digital

Figure 4: MLO view comparison between Film and Digital Mammography. Images adapted from CBIS-DDSM and RSNA respectively.

Dataset Name	Source	Year	Image Resolution	# Patients	# Missing Labels	# Benign	# Malignant	# Total	Data Size (GB)
CBIS-DDSM (Lee et al., 2017)	USA	2017	Fig. 5	1,566	—	1,429	1,457	2,886	163.6
VinDr (Nguyen et al., 2023)	Vietnam	2022	(3,580 x 2,685) ± (0, 0)	5,000	—	18,357	2,129	20,000	337.8
CMMD (Cai et al., 2023)	China	2023	(2,294 x 1,914) ± (0, 0)	1,775	3,330	556	1,316	5,202	22.9
RSNA (Carr et al., 2022)	USA	2023	(3,526 x 2,082) ± (0, 0)	11,900	—	53,548	1,158	54,700	314.72

Table 3: Summary of publicly available datasets related to breast cancer

Properties	CBIS-DDSM	CMMD	VinDr	RSNA
Mammography Type	Film	Digital	Digital	Digital
Anonymised Patient ID	✓	✓	✓	✓
Image Laterality	✓	✓	✓	✓
Image Dimensions	✓	✓	✓	✓
Diagnosis	✓	✓	✓	✓
Lesion Type	✓	✓	✓	
Age		✓	✓	✓
Image View	✓		✓	✓
Breast Density (BI-RADS)	✓		✓	✓
Training/Test Split	✓		✓	✓
ROI	✓		✓	
Scanner		✓	✓	

Table 4: Dataset Properties found in DICOM files and metadata

4 Related Work

In the field of breast cancer diagnosis and classification, significant advancements have been made through the application of deep learning techniques. From image preprocessing to CNNs used, these papers laid down the foundations for future research in this field.

Similar outcomes were achieved in other medical domains and modalities such as classifying patients with autism spectrum disorders via MRI images (Li et al., 2020). Screening for COVID-19 from chest X-ray images was also found to perform well in an FL environment as seen in Feki et al. (2021), L. Zhang et al. (2021) and Abdul Salam et al. (2021). As such, we aim to explore the usage of FL for diagnosing breast cancer with mammograms.

4.1 Mammography Segmentation

A lesion only makes up a small area of the entire mammogram image, but determines the image's pathology. This is observed from a radiologist's workflow when he looks at a mammogram, scanning the entire image to find suspicious areas that may be lesions. This also helps to reduce manual work of annotating future datasets, that is usually done by radiologists. However, when scaled up to a large dataset, annotating radiologists will be prone to fatigue and human errors. As such, research has been looking into automating the process of finding lesions in a mammogram.

Singh and Gupta (2015) applied image processing techniques like thresholding and averaging and were successful in locating lesions in the test images. However, the manual tuning of parameters of the processing techniques is not feasible for large dataset, as there is no one-size-fits-all parameter.

Many have hence explored using CNNs for the segmentation task, particularly U-Net, which was specifically developed for medical imaging segmentation (Ronneberger et al., 2015). Zeiser et al. (2020) proposed using U-Net to monitor the progress of lesion, and their best model was able to achieve 0.8595 accuracy, 0.8640 AUC, 0.9232 sensitivity (recall), 0.8047 specificity and Dice Index of 0.7939. Hossain (2022) also applied U-Net to segment mammograms, getting 0.982 average accuracy, 0.985 F1 Score, 0.978 Dice score. Byra et al. (2020) developed a novel network architecture based on U-Net to identify masses in breast ultrasounds, achieving a mean Dice score of 0.826, outperforming a regular U-Net that has a Dice score of 0.778. Adoui et al. (2019) used SegNet and U-Net on MRI for breast cancer, achieving mean intersection over union (IoU) of 0.6888 and 0.7614 respectively. Baccouche et al. (2021) integrated two U-Net networks together to create a novel architecture called "Connected U-Nets". They also used cycle-consistent Generative Adversarial Network (CycleGAN) model to augment and enhance unrelated datasets, creating synthetic data and then conducted experiments with the additional data. The proposed network attained a Dice score of 0.8952, 0.9528, and 0.9588 and IoU score of 0.8002, 0.9103, and 0.9227 for CBIS-DDSM, INbreast and their private dataset respectively.

4.2 Mammography Classification

Given that mammograms are one of the few medical image modalities with high resolutions, it is necessary to apply appropriate preprocessing methods without losing important information. Methods like applying denoising filters, contrast enhancement using histogram equalization, removal of pectoral muscle and artifact removal were commonly suggested in these papers Yu et al. (2023), Lbachir et al. (2020), Tripathy and Swarnkar (2020), and were found to improve performance.

The works of Falconi et al. (2020) provided insights into the effectiveness of transfer learning and fine-tuning in mammogram abnormalities classification using the CBIS-DDSM database. Falconi's work emphasized the use of transfer learning and fine-tuning of pretrained models to improve performance on mammography classification. When data augmentation was introduced, they were able to achieve an area under curve (AUC) value of 0.844 as a result. Additionally, Lbachir et al. (2020) introduced an automatic computer-aided diagnosis system for mass detection and classification in mammography, employing ResNet50 to achieve an accuracy of 0.904. Salama et al. (2020) delved into novel classification frameworks, exploring the utilization of support vector machines (SVM) as a classifier and attained 0.9798 accuracy, 0.9846 AUC, 0.9651 precision, 0.9597 F1 score on the DDSM dataset. Another study done by Salama et al. extended the exploration by implementing segmentation with a modified U-Net model and InceptionV3 model, achieving 0.9887 accuracy, 0.9888 AUC, 0.9879 precision, 0.9799 F1 score on the same dataset (Salama & Aly, 2021).

Petrini et al. (2022) achieved an accuracy of 0.85 and a remarkable AUC of 0.934 by taking into account both views of a mammography scan. While considering the breast's BI-RADS classification and the views of each mammogram, Wu et al. (2020) was able to achieve an AUC of 0.895 using the NYU Breast Cancer Screening Dataset (Wu et al., 2023).

4.3 Patch Classifiers

In various studies, the utilization of whole mammographic images and their ROIs as the training and testing dataset has been observed, while others opt for employing patch classifiers. A patch classifier is a technique that breaks down the image into smaller sections or patches to train the network, allowing for a more detailed analysis of specific regions. By focusing on these smaller segments, the patch classifier enables the model to detect subtle nuances or patterns as demonstrated by Yu et al. (2020). This is explained by Shen et al. (2019) as they raised an observation of a malignant lesion making a small region of the entire mammogram, but being the main factor in determining the mammogram's diagnosis. Therefore, this sliding window approach tends to capture localized features within the image, aiding in a more nuanced and precise classification, especially beneficial in complex or detailed mammographic images. However, it has its drawbacks such as not recognising the locality information between patches, and also doubling the computational resources for every patch as mentioned in Quintana et al. (2023).

4.4 Federated Learning

As for the application of breast cancer classification in an FL setting, there has been remarkable studies as well. Tan et al. (2023) developed a novel solution in classifying breast cancer by extracting data features from ROI images, and techniques to address imbalanced datasets in an FL environment. They achieved 0.9279 accuracy and 0.9904 AUC using balanced data, and 0.8933 accuracy and 0.9699 AUC using imbalanced data, with FeAvg-CNN and MobileNet. Roth et al. (2020) explored the possibility of integrating FL in breast cancer classification on real-world datasets, achieving 6.3% average improvement using the models trained in FL compared to the models that trained using only its own institute's local data. Jim´enez-S ´anchez et al. (2023) introduced a novel memory-aware curriculum learning approach within federated adversarial learning to enhance multi-site breast cancer classification. By prioritizing forgotten samples in local training updates, combined with domain adaptation, they improved AUC and PRAUC (Precision-Recall AUC) by an average of 5% and 6% respectively, across three clinical datasets from different vendors.

Table 5 summarizes all papers mentioned in this section, including the datasets used and performances of the models. These studies not only contribute to the growing body of knowledge in breast cancer diagnosis but also provide valuable insights into the choice of network architectures and data augmentation techniques.

Table 5: Summary of state-of-the-art (SOTA) methods used in breast cancer diagnosis
ACC: Accuracy. AUC: Area under Curve. P: Precision. R: Recall. F1: F1 Score. IoU: Intersection over Union.

Author	Contribution	Research Gaps	Image Dataset	Model	Performance
Segmentation					
Singh and Gupta, 2015	Using thresholding and averaging to locate lesions in mammograms, reducing time needed for radiologists to diagnose a mammogram	Manual tuning of image processing parameters, making it not feasible for larger datasets	Not mentioned	—	~4.20s
Zeiser et al., 2020	Using U-Net to monitor progress of lesion	Network can be made more robust to a mammogram's breast density, so that it can be detect lesions even with different densities. This can be done by taking in the density into account when training the network	Mass Whole Images from DDSM	U-Net	ACC: 0.8595 AUC: 0.8640 R: 0.9232 Specificity: 0.8047 Dice: 0.7939
Hossain, 2022	Using a novel U-Net-based architecture with less number of parameters to detect calcifications	Performance of the same network was found to be poor on MIAS dataset. Possible reasons were due to the dataset being of lower contrast compared to the DDSM dataset, resulting in lower intensity difference between the calcification and non-calcification regions	MLO Whole Images from DDSM	U-Net	ACC: 0.982 F1: 0.985 Dice: 0.978

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Author	Contribution	Research Gaps	Image Dataset	Model	Performance
Byra et al., 2020	Developed a novel architecture based on U-Net to identify masses in ultrasounds	Small ultrasound dataset of 882 images, which were annotated by different radiologists. This may lead to variance in labelling of ROIs	ROIs from UDIAT (Yap et al., 2017), OASBUD (Piotrzkowska-Wróblewska et al., 2017), BUSI (Al-Dhabyani et al., 2020)	Novel SK-U-Net	Dice: 0.826
Adoui et al., 2019	Compared performance of SegNet and U-Net on MRI for breast cancer	Small dataset of 86 MRI volumes (Train: 60; Test: 26)	Whole Images from collaborator (Jules Bordet Institute)	SegNet, U-Net	IoU (SegNet): 0.6888 IoU (U-Net): 0.7614
Baccouche et al., 2021	Developed a new architecture to segment masses in mammograms using methods like attention mechanism, residual blocks etc. Also used cycleGAN model to create synthetic images to increase size of training data	Size of the complex networks means a huge number of parameters, which causes long inference times for segmenting new images	Whole Images from CBIS-DDSM, INbreast, Private dataset	Novel Connected U-Nets architecture	Dice (CBIS-DDSM): 0.8952 Dice (INbreast): 0.9528 Dice (Private): 0.9588 IoU (CBIS-DDSM): 0.8002 IoU (INbreast): 0.9103 IoU (Private): 0.9227

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Author	Contribution	Research Gaps	Image Dataset	Model	Performance
Classification					
Falconi et al., 2020	Application of deep learning networks onto breast cancer classification	—	ROIs from CBIS-DDSM	VGG16	AUC: 0.844
Lbachir et al., 2020	Developing a computer-aided diagnosis (CADx) system that segments lesions in mammograms and classify it	Network was not able to segment spiculation in the masses' outlines, which is crucial in determining whether the lesion is malignant or not	ROIs from MIAS, CBIS-DDSM	SVM Classifie r	ACC: 0.904
Salama et al., 2020	Combining ResNet50 net- work with a SVM classifier	No source code	Whole Im- ages from MIAS, DDSM, CBIS-DDSM	ResNet5 0 hy- bridised with SVM	ACC: 0.9798 AUC: 0.9846 P: 0.9651 F1: 0.9597
Salama and Aly, 2021	Added ROI seg mentation network to gain more spatial information around ROIs, which improved performance to- gether with data augmentation and the use of both MLO and CC views	No source code	ROIs from MIAS, DDSM, CBIS-DDSM	U-Net model + Incep- tionV3	ACC: 0.9887 AUC: 0.9888 P: 0.9879 F1: 0.9799
Petrini et al., 2022	Apply transfer learning from single-view whole image classifier to a two-view whole image classifier	Include BI-RADS label, and other patient details to improve network's classification	Whole Im- ages from CBIS-DDSM	EfficientNet	ACC: 0.85 AUC: 0.934 (5-fold cross validation) AUC: 0.8483 (original dataset division)

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Table 5 – continued from previous page

Author	Contribution	Research Gaps	Image Dataset	Model	Performance
Wu et al., 2020	Pretraining using BI-RADS classification; Network that is able to learn at pixel-level and breast-level; Integration of multiple input views	—	Whole Images from NYU Breast Cancer Screening Dataset	ResNet22	AUC: 0.895

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Table 5 – continued from previous page

Author	Contribution	Research Gaps	Image Dataset	Model	Performance
Federated Learning					
Tan et al., 2023	Presentation of experimental results from diverse deep learning, transfer learning, and FL models with both balanced and imbalanced mammography datasets	Required long training time using FL compared to centralized training; CNNs required data preprocessing methods to be applied to the mammogram prior to classifying it	ROIs from DDSM, CBIS-DDSM	MobileNet + FeAvgCNN	Balanced Data: ACC: 0.9279 AUC: 0.9904 Imbalanced Data: ACC: 0.8933 AUC: 0.9699
Roth et al., 2020	Implementation of medical imaging classification models in a real-world collaborative setting, predicting mammogram's BIRADS label	No source code and research was done on private data, thus only showing a successful application of the thesis, but not allowing further continuation based on this research	Whole Images from Private Dataset	DenseNet-121 + FedAvg	—
Jiménez-Sánchez et al., 2023	Prioritizes forgotten or inconsistent samples in training by penalizing inconsistent predictions.	—	Whole Images from Hologic (Private), GE (Private) and Siemens INBreast (Public)	ResNet22 + FedAlignCL	AUC: 0.79 PRAUC: 0.82

5 Methodology

5.1 Data Analysis

The fundamental aim of this study is to classify a mammogram benign or malignant. Therefore, class distribution of the datasets used in this study needs to be accounted for, to make sure the data is not skewed to one pathology. Table 3 has previously showed the class distribution of the diagnosis for each dataset. CBIS-DDSM was used to train the centralized model because it was the most balanced dataset among all that we have found available. This ensures that performance is not affected by the class imbalance.

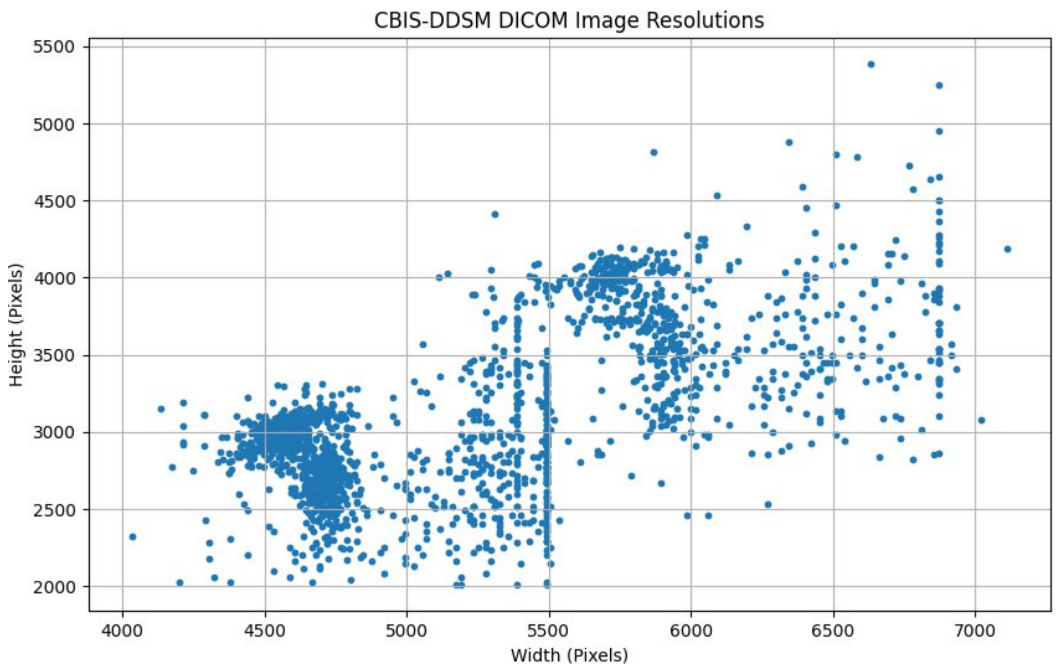


Figure 5: Resolutions of original images in CBIS-DDSM dataset. Average Resolution: (5265.52, 3132.35) ± (668.81, 537.54)

The CBIS-DDSM dataset presents a diverse array of image resolutions. Figure 5 provides a visual representation of the spectrum of resolutions in the dataset. To gain a deeper insight into this variability, we calculated the average resolution along with the standard deviation of both width and height across the dataset.

5.2 Data Preprocessing

As mentioned in Section 4, previous papers have highlighted the importance of data preprocessing of mammograms to allow ML models to learn more effectively. Unlike some other medical images, mammograms exhibit distinct characteristics such as exceptionally high contrast, fine-grained details, and high resolution. These characteristics can both enhance and complicate the interpretability of mammograms. Preprocessing steps are, therefore, indispensable in mitigating potential challenges and optimizing the diagnostic utility of these images.

By addressing issues like noise reduction, contrast enhancement, and image standardization, preprocessing ensures that mammograms are primed for accurate and consistent interpretation. Moreover, given the crucial role of mammography in early breast cancer detection, the importance of preprocessing extends beyond mere image enhancement; it significantly impacts patient outcomes by reducing false positives, improving lesion detectability, and facilitating the work of radiologists. As a result, the rigorous preprocessing of mammograms remains an indispensable component of the breast cancer screening and diagnostic pipeline, ultimately contributing to improved healthcare and enhanced quality of life for individuals worldwide.

The preprocessing methods used in this study were inspired from previous studies, and they include extraction of ROI-specific regions from the original image, removal of artifacts (Tripathy & Swarnkar, 2020), applying denoising filters and histogram equalization (Lbachir et al., 2017), and pectoral muscle removal (Vagssa et al., 2020).

5.3 Federated Learning Frameworks

When considering how we could implement FL into our experiments, several frameworks were studied for their advantages and disadvantages, including OpenFL (Foley et al., 2022), Tensorflow Federated (TFF) (Bonawitz et al., 2019) and PySyft (Ryffel et al., 2018). We decided on using Flower as our main FL framework (Beutel et al., 2022) because it delivers a stable foundation for the fundamental components of a FL system. It also allows researchers to explore and put fresh concepts into practical implementations on top of the existing infrastructure. Given Flower's detailed documentation and straightforward setup, it was decided that Flower was used as the FL framework.

5.4 Performance Metrics

The de facto metrics for training machine learning models are accuracy and loss. While we used these metrics to make sure the model is learning well from the data and not overfitting or underfitting, the nature of this task requires us to consider using other metrics such as precision, recall and F1 score. This consideration was also supported by other papers in similar domains (Das et al., 2023), (J. Zhang et al., 2019). Reasons for the usage of precision and recall include the reduction of false positives (which can lead to unnecessary medical examinations and patient anxiety) and false negatives (which can lead to late treatments of the disease) respectively.

As such, the metrics used in this study include accuracy, precision, recall (sensitivity), F1-score, area under the receiver operating characteristic curve (AUC), and area under the precision-recall curve (PRAUC). Each metric holds a specific significance in evaluating the model's performance.

Accuracy signifies the proportion of correct predictions over all predictions, calculated as

$$\frac{TP + TN}{TP + TN + FP + FN}$$

where TP denotes true positives, TN signifies true negatives, FP represents false positives, and FN indicates false negatives.

Precision reflects the ratio of correctly identified positive predictions to all predicted positives, computed
as

$$\frac{TP}{TP + FP}$$

Recall, or sensitivity, measures the proportion of correctly identified positive predictions out of all actual positives, expressed
as

$$\frac{TP}{TP + FN}$$

F1-score is the harmonic mean of precision and recall, given by

$$\frac{2 \times precision \times recall}{precision + recall}$$

Additionally, AUC and PRAUC evaluate the model's ability to differentiate between classes, with AUC representing the area under the ROC curve, and PRAUC denoting the area under the precision-recall curve. These metrics collectively provide a comprehensive assessment of the model's performance in breast cancer detection.

6 Experiments

A set of experiments was done to design a baseline model. We compared different networks and assessed the impact of data preprocessing and augmentation. We also explored various aspects of model performance, from metrics selection to dataset balancing. The following sections will explain more about the experiments, the hypothesis and the results that was obtained.

6.1 Experimental Setup

6.1.1 Data Loading Efficiency from DICOM to JPEG

To speed up the process of training and validating the model on the data, we explored the conversion of the DICOM files to JPEG file, a common picture file format, thereby loading pixel data from the JPEG file instead of the DICOM file. It was found that a one-off conversion of all the DICOM files to JPEG gives rise to a speedup in data loading as shown in Table 6.

	DICOM	**JPEG**
Training Time	790s	510s
Validation Time	320s	200s

Table 6: Average time taken for loading data from DICOM/JPEG files

6.1.2 Comparing Image Stacking Methods for Speedup

Mammograms are grayscale in nature, which means they only have one image channel. This is incompatible with many SOTA models because they are trained on coloured images, and thus require 3-channel (RGB) inputs. Two methods were found to enable the model to train on the grayscale mammographic images:

1. Duplicate the image's single-channel pixel data on 3 channels
2. Use torchvision.transforms library (ToGrayscale()) to convert the image to RGB image

	Time Taken To Load Image
Stacking using Numpy	0.052s
ToGrayscale()	0.12s

Table 7: Time taken for converting grayscale image to RGB

As seen in Table 7, it is less time-consuming to stack the image using Numpy library, which decreases the overall time required to load data from the JPEG files.

6.1.3 Using *num_workers* relative to number of CPU cores

The choice of the *num_workers* parameter in PyTorch's DataLoader object is directly linked to the available computational resources, particularly the number of CPU cores or processors at the disposal of the system. The *num workers* value signifies the degree of parallelism during data loading, enabling multiple CPU cores to work concurrently in fetching and preparing data for the neural network. To ensure efficient resource utilization, setting *num workers* relative to the number of CPU counts is a sensible approach. This alignment optimizes data loading, mitigating potential bottlenecks, and ensures that the CPU cores are utilized effectively. By striking this balance between *num workers* and CPU counts, one can harness the full potential of the hardware and expedite the data preparation phase, thereby enhancing the overall efficiency and training performance of deep learning models.

6.1.4 Non-Blocking for Speedup

The choice to employ a non-blocking approach in data loading hinges on optimizing system resources and enhancing efficiency. Non-blocking operations allow multiple tasks to run concurrently, eliminating wait times and avoiding resource under utilization. This approach is particularly advantageous when ample computational resources are available.

Listing 1: Non-Blocking in PyTorch

```
data = data.to(device, non_blocking=True)
```

6.1.5 Optimizing Hyperparameters (Learning Rate (LR), Batch Size, Optimizer)

When doing image classification on the CBIS-DDSM dataset, other papers that did this similar task were studied and analysed to see how they got their results. Table 5 inspired the networks and parameters that was used to create a baseline model.

6.1.6 Weight Decay

One of the methods to reduce overfitting is to use weight decay. Weight decay addresses this issue by adding a regularization term to the loss function during training, encouraging the model to maintain smaller weights for its parameters (Krogh & Hertz, 1991). This regularization mitigates the risk of overfitting by effectively penalizing overly complex models. By imposing a constraint on the magnitude of weight values, weight decay promotes a more generalizable and robust model, striking a crucial balance between fitting the training data and avoiding the pitfalls of overfitting. In this way, weight decay provides a vital mechanism to enhance the performance and adaptability of deep learning models, fostering their ability to generalize well to unseen data.

6.1.7 GradCAM

GradCAM, short for Gradient-weighted Class Activation Mapping, is a sophisticated technique that empowers deep learning models with a unique and interpretable dimension (Selvaraju et al., 2019). By capturing the network's attention at the last few convolutional layers, GradCAM highlights which regions of an image are crucial for its predictions, shedding light on the inner workings of complex neural architectures. This visualization tool offers a critical advantage in understanding the model's decision-making process, going beyond the traditional "black box" nature of deep learning. With GradCAM, we can gain insights on how a model made the decision to classify a picture.

6.2 Centralized Training

6.2.1 Comparing Results for Different Networks

Multiple SOTA models were tested against each other on CBIS-DDSM dataset to see which produced the best performance. We used the original train/test split that came originally in the dataset. The networks are trained and validated using ROI images only since ROI image classifiers are found to give better performance than whole image classifiers in past research. Only mass images were used for the experiments. As this is the first set of experiments, no data augmentation was applied. Also, weights pretrained on ImageNet were used for all the networks. Networks were frozen, except the final convolutional layer and the classifier layer. This is to allow the network to learn and adapt to mammographic images. These are the parameters used in this section:

- **Learning Rate:** 10^{-4}
- **Batch Size:** 512
- **Epochs:** 100
- **Optimizer:** Adam
- **Loss Function:** Binary Cross Entropy Loss
- **Weight Decay:** 10^{-2}
- **Input Resolution:** (224, 224)
- **Train / Validation Ratio:** 0.8 / 0.2 (Refer to Table 8)

Data	Benign	Malignant
Training	542	512
Validation	139	125
Test	231	147

Table 8: Class Distribution in CBIS-DDSM (Mass) Dataset

Results from this experiment can be found in Table 9. While EfficientNet-b0 definitely performed poorly with an extremely high validation loss, performance of other networks are comparable based on statistics. However, these numbers do not paint the full picture, as it does not reflect any possible overfitting or underfitting trends. Upon investigating the loss performance in Figure 6, it was found that ResNet50 gave the least overfitting curve, and given that it also gave comparable results to the other four networks, it was decided to be used as the network for the rest of the study.

Network	Parameters		Training		Validation						
	Trainable	Total	Accuracy	Loss	Accuracy	Loss	Precision	Recall	F1	ROC	PRROC
MobileNet v2	1,084,481	2,896,193	0.8880	0.3816	0.6705	0.7001	0.6418	0.688	0.6641	0.6709	0.7558
EfficientNet_b0	1,081,921	4,679,869	0.8065	0.4609	0.5530	42.3495	0.7333	0.088	0.1571	0.6378	0.4182
DenseNet121	707,329	7,495,105	1.0	0.0219	0.6705	0.9399	0.6462	0.672	0.6588	0.6701	0.7462
Xception	4,225,089	21,872,489	0.9915	0.1344	0.6894	0.6788	0.7009	0.6	0.6466	0.6912	0.7111
ResNet50	2,118,209	24,573,569	0.7789	0.5129	0.6818	0.5767	0.6454	0.728	0.6842	0.6845	0.7814

Table 9: Performance of SOTA models

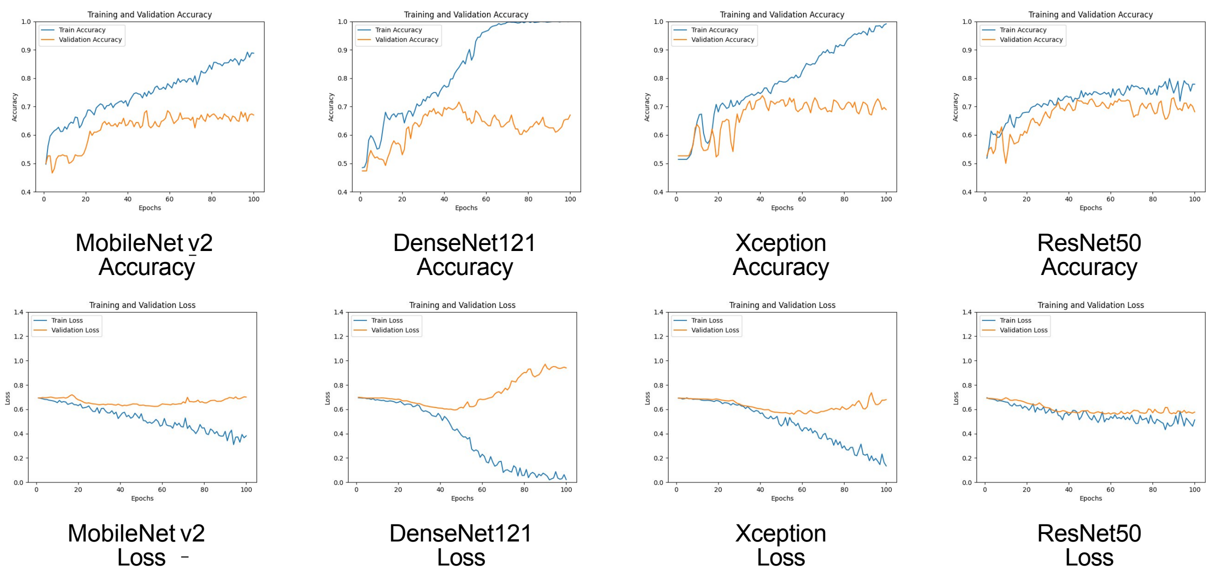


Figure 6: Graphical performance of SOTA models in Table 9

6.2.2 Transfer Learning

Transfer learning is often done from ImageNet, but we tried doing it from medical images. We expected improved performance of the model since similar information may be extracted from images from the same domain. This was done using MedMNIST (Yang et al., 2023), testing with both the weights trained from ChestMNIST and PneumoniaMNIST datasets. These datasets were the most similar to mammograms because they are all chest X-rays, except that the latter is specifically for the breast.

Unfortunately, results were not as what we expected, doing significant harm to the model's performance as seen in Table 10. This drop in performance was also seen in (Wen et al., 2021)'s work, and Wen et al. mentioned that this could be due to the lack of morphological awareness in the model. It is also probable that the number of images in ImageNet being more than 126 times of that in the ChestMNIST dataset, and 2,400 times of that in PneumoniaMNIST dataset. Large amounts of data allows for better performance in machine learning models (Sarker, 2021). As such, ImageNet weights were used for the rest of the study.

6.2.3 Data Selection

CBIS-DDSM dataset provides both ROI and whole images. We experimented with a mixture of the two types to see if it improves performance. Table 11 presents the outcomes of these experiments using ResNet50 network.

The results indicate superior validation performance when both training and validation sets are of the same image type.

Metric	ImageNet	ChestMNIST	PneumoniaMNIST
# Images	14,197,122	112,120	5,856
Train Accuracy	0.7789	0.5522	0.5266
Train Loss	0.5129	0.6936	0.6901
Val. Accuracy	0.6818	0.5114	0.5265
Val. Loss	0.5767	0.6856	0.6862
Val. Precision	0.6454	0.4688	0.5
Val. Recall	0.728	0.24	0.416
Val. F1	0.6842	0.3175	0.4541
Val. ROC	0.6845	0.4969	0.5219
Val. PRROC	0.7814	0.4188	0.5565

Table 10: Performance metrics for using ImageNet, ChestMNIST, PneumoniaMNIST weights

Training Set	Validation Set	Training		Validation						
		Accuracy	Loss	Accuracy	Loss	Precision	Recall	F1	ROC	PRROC
ROI	ROI	0.7789	0.5129	0.6818	0.5767	0.6454	0.728	0.6842	0.6845	0.7814
ROI and Whole images	Whole images	0.7709	0.5028	0.6174	0.7034	0.5833	0.672	0.6245	0.6208	0.7413
ROI	Whole images	0.7799	0.5024	0.5076	0.9383	0.4390	0.144	0.2169	0.4796	0.3351
Whole images	Whole images	0.7761	0.5053	0.6477	0.6640	0.5941	0.808	0.6847	0.6694	0.8317

Table 11: Model performance on non-preprocessed images

Figures 7 illustrate the accuracies, loss, and ROC curves for the non-preprocessed images. Both training and validation sets were using the same image type in these two experiments.

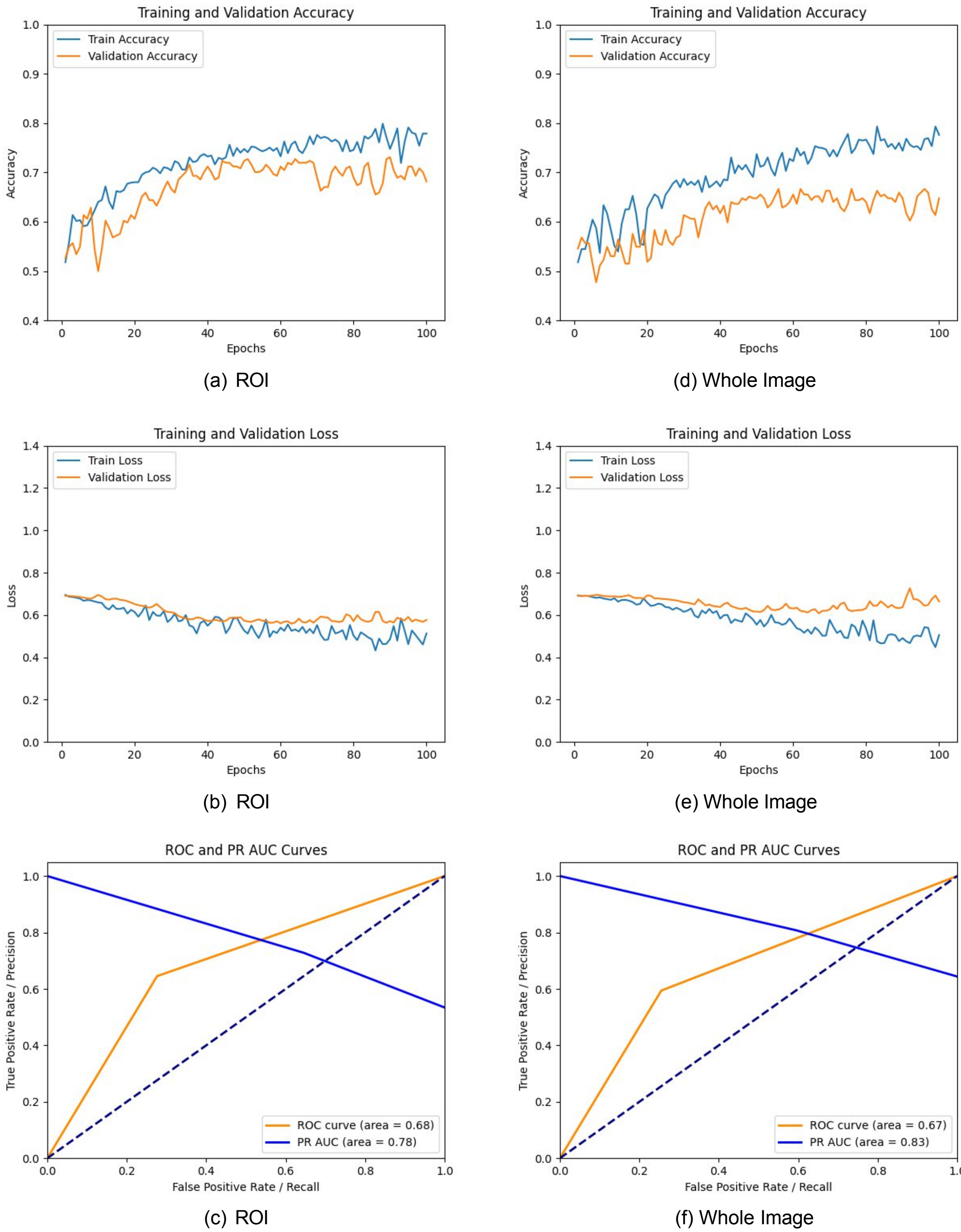
While the performance of all-ROIs and all-whole-images are comparable, we decided to use the all-ROIs approach for the rest of the study because it performed better generally, except F1-score being comparable and PRROC being worse.

6.2.4 Data Augmentation

Inspired by multiple previous studies, common data augmentation methods used for mammograms are random rotations, random vertical and horizontal flipping, and occasionally scaling. Table 12 shows the effect of these augmentations on the performance.

After some experimentation, it was found that using a combination of the data augmentation methods gave the lowest validation loss. We want to keep the validation loss low to make sure the model remains robust to future unseen data. The combination of methods is as follows:

- Random Vertical Flip
- Random Horizontal Flip
- Random Scaling (between 1x to 1.2x zoom)



(a) ROI (d) Whole Image

(b) ROI (e) Whole Image

(c) ROI (f) Whole Image

Figure 7: Performance comparison between ROI and whole images

6.3 Federated Learning

When training models in FL, a new variable known as communication rounds is introduced. One communication round consists of the central server distributing the central model to the institutions, and then

Data Augmentation	Training		Validation	
	Accuracy	Loss	Accuracy	Loss
—	0.9383	0.2430	0.7273	0.7346
Random Vertical Flip	0.8359	0.3863	0.6553	0.6967
Random Horizontal Flip	0.6964	0.5867	0.6477	0.6061
Random Rotation 90°	0.6774	0.5722	0.6439	0.6241
Random Scaling (between 1x to 1.2x zoom)	0.7419	0.5378	0.6970	0.5707
Combination	0.7258	0.5504	0.7273	0.5431

Table 12: Effect of data augmentation techniques

receiving all the trained models from the institution, and aggregating it according to the decided aggregation algorithm. Common aggregation algorithms include FedAvg (McMahan et al., 2023b) which does average aggregation, FedOPT (Reddi et al., 2020) for weighted aggregation, FedBoost for ensembled aggregation (Hamer et al., 2020), just to name a few.

For our experiments, FedAvg was used. Inspired by Tan et al., 2023, we also used 3 training epochs at each institution. As for communication rounds, we tried two variations, 20 and 60 (as mentioned in said paper).

We used 4 datasets, namely CBIS-DDSM, CMMD, RSNA and VinDr. As seen in Table 3, RSNA and VinDr's datasets are about 10x of the size of CBIS-DDSM and CMMD. We intend to undersample the former two according to the minority class, so that the size of all four datasets will be roughly the same. This will help with FL training times as there will not be any bottlenecking of individual training times.

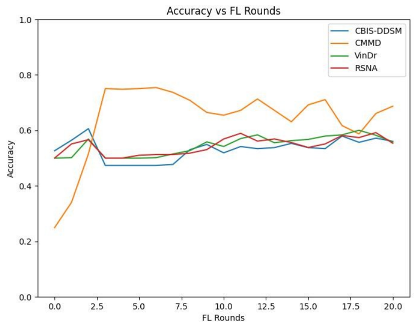
6.3.1 Difference in Communication Rounds

Here are the average performance statistics of the FL model for each dataset when ran under different communication rounds. Table 13 and Figure 8 and 9 shows the performance for 20 vs 60 communication rounds.

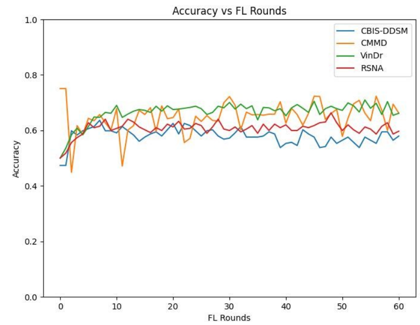
Metrics	20 Rounds	60 Rounds
Accuracy	0.5665	0.6258
Loss	0.6848	0.8487
Precision	0.5831	0.6503
Recall	0.7767	0.7248
F1 Score	0.6240	0.6691
AUC	0.5577	0.5951
PRAUC	0.7685	0.7630

Table 13: Average metrics for 20 rounds and 60 rounds

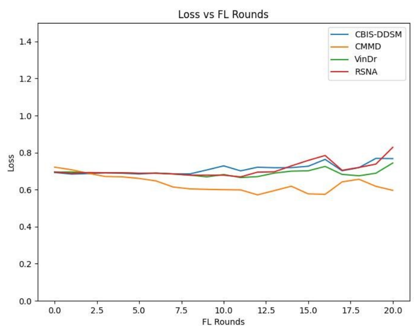
Performance is noticeably better when the FL model is able to train for longer communication rounds, but higher loss is also inevitable due to the model being exposed to more variety of images every round. We saw an improvement in F1 score and AUC with more communication rounds, which are important in the medical domain for accurate diagnosis and reduction of misdiagnosis.



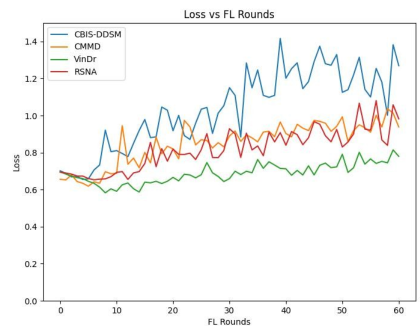
(a) FL Accuracy (20 rounds)



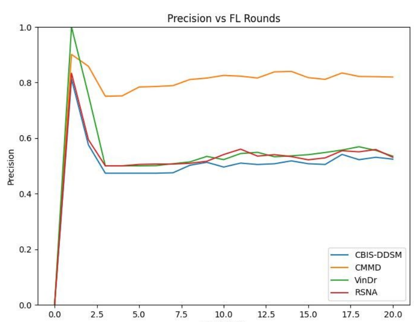
(e) FL Accuracy (60 rounds)



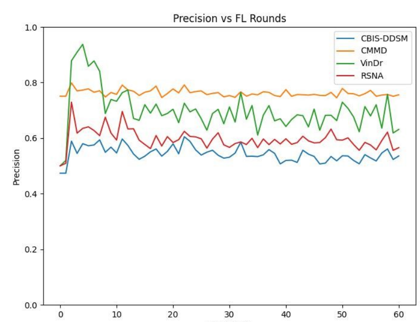
(b) FL Loss (20 rounds)



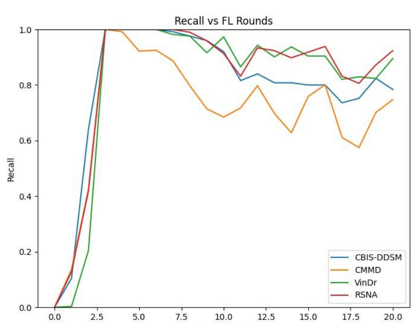
(f) FL Loss (60 rounds)



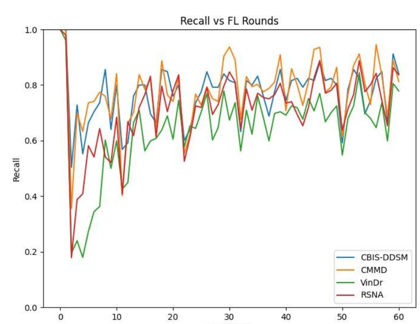
(c) FL Precision (20 rounds)



(g) FL Precision (60 rounds)

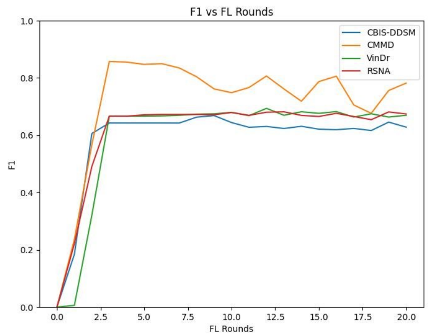


(d) FL Recall (20 rounds)

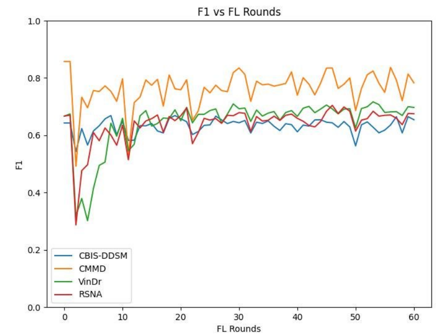


(h) FL Recall (60 rounds)

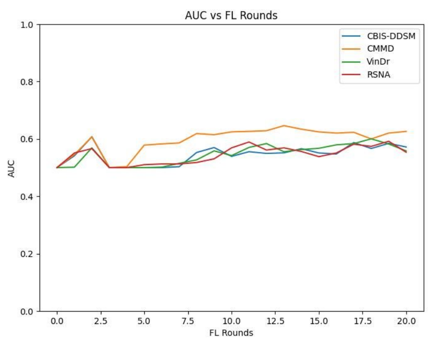
Figure 8: FL Performance (20 vs 60 rounds)



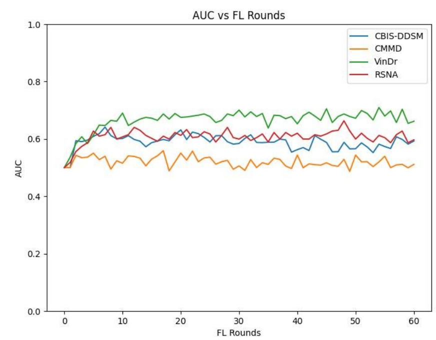
(a) FL F1 (20 rounds)



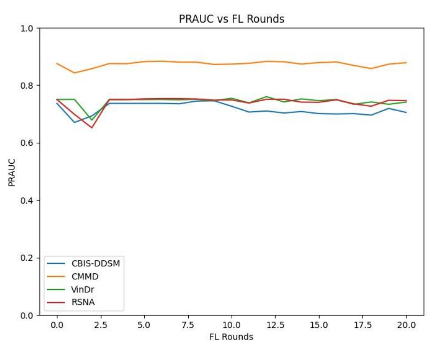
(d) FL F1 (60 rounds)



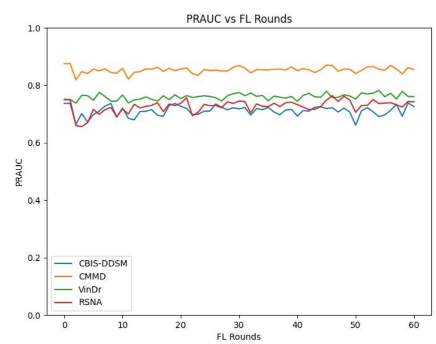
(b) FL AUC (20 rounds)



(e) FL AUC (60 rounds)



(c) FL PRAUC (20 rounds)



(f) FL PRAUC (60 rounds)

Figure 9: FL Performance (20 vs 60 rounds) [continued]

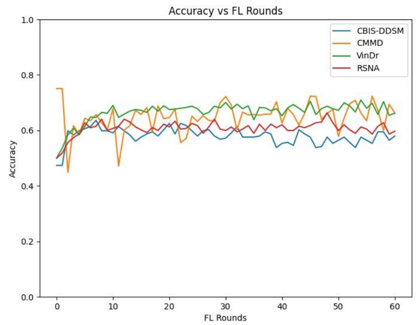
6.3.2 ImageNet vs Pretrained ROI Weights

We also tried fine-tuning a ResNet50 network on ROI images from CBIS-DDSM, and subsequently using this model as the FL model. Since more communication rounds were found to be better from the previous section, we used 60 communication rounds in this experiment, making sure the starting weights of the FL model are the only variable. Table 14 and Figure 10 and 11 shows the performance found in this experiment.

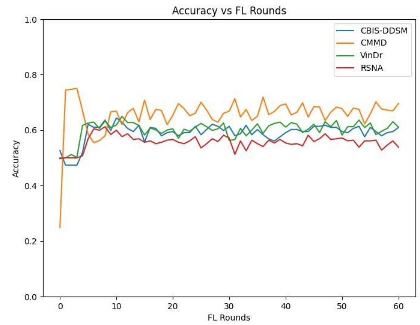
Metrics	ImageNet Weights	Fine-tuned Weights
Accuracy	0.6258	0.6021
Loss	0.8487	0.7502
Precision	0.6503	0.6135
Recall	0.7248	0.7539
F1 Score	0.6691	0.6632
AUC	0.5951	0.5875
PRAUC	0.7630	0.7591

Table 14: Average metrics for different weights

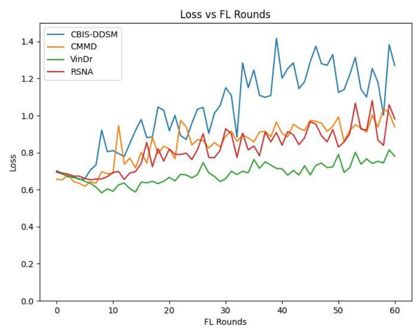
While the fine-tuned weights did not perform as well, it has shown potential as it differs from the other model marginally. With its accuracy having a difference of 2% and precision having that of about 4%, it made up for it in metrics such as loss (+9%), recall (+3%), and F1 score, AUC and PRAUC being within 1% of its competition.



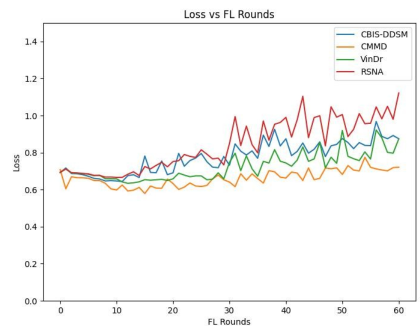
(a) FL Accuracy (ImageNet weights)



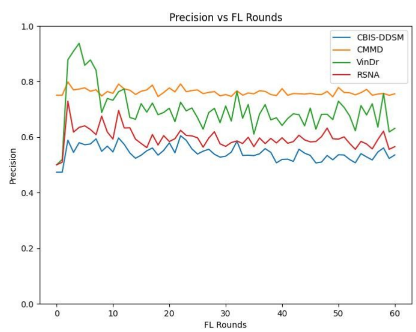
(e) FL Accuracy (Fine-tuned weights)



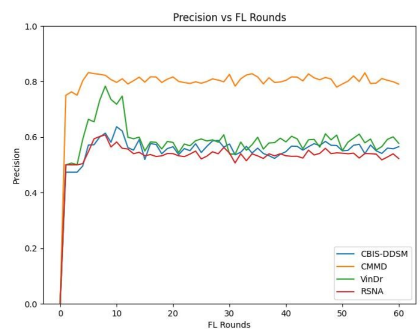
(b) FL Loss (ImageNet weights)



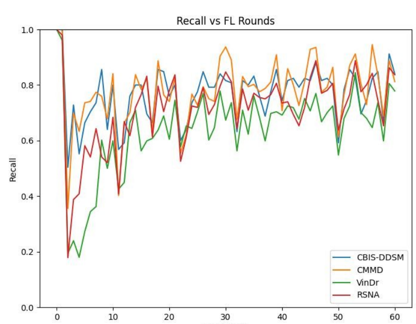
(f) FL Loss (Fine-tuned weights)



(c) FL Precision (ImageNet weights)



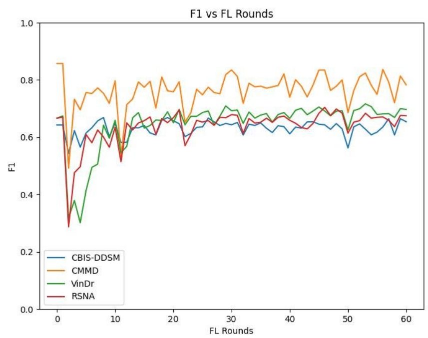
(g) FL Precision (Fine-tuned weights)



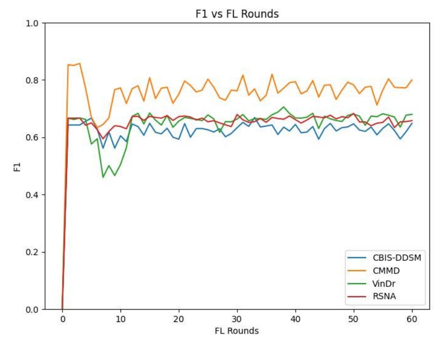
(d) FL Recall (ImageNet weights)

(h) FL Recall (Fine-tuned weights)

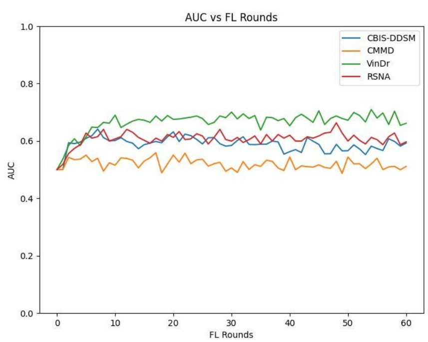
Figure 10: FL Performance (ImageNet vs Fine-tuned weights)



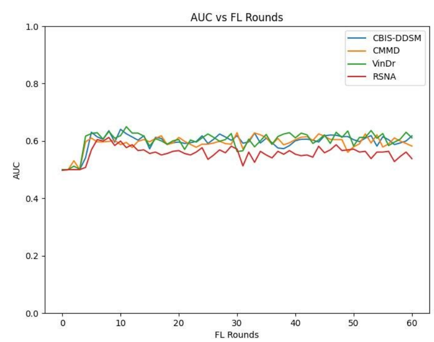
(a) FL F1 (ImageNet weights)



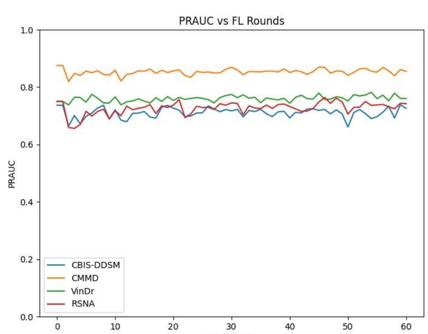
(d) FL F1 (Fine-tuned weights)



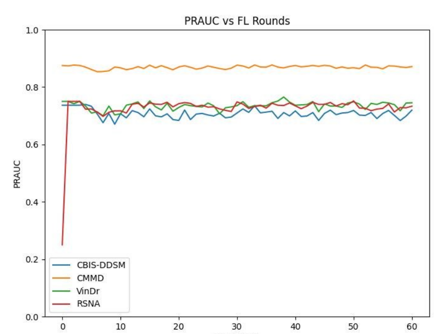
(b) FL AUC (ImageNet weights)



(e) FL AUC (Fine-tuned weights)



(c) FL PRAUC (ImageNet weights)



(f) FL PRAUC (Fine-tuned weights)

Figure 11: FL Performance (ImageNet vs Fine-tuned weights) [continued]

7 Conclusion

7.1 Summary

This study looked into modern classification methods for mammograms and their application within the emerging field of FL. Several publicly available datasets were studied, such as CBIS-DDSM, CMMD, VinDr, and RSNA, and the vastly different datasets allow for models to become more robust. Incorporating multiple datasets sourced from diverse geographical regions introduces inherent variability in the geographic origin of patients and how a scan looks. This is a common problem that exists as non-IID (Independent and Identically Distributed) data in FL, which causes performance in FL to be worse due to multiple factors unrelated to the disease itself.

Our objective, when using such different datasets, was to emulate the FL environment authentically and include these inherent challenges and differences. When developing such technology for medical purposes, it is also important to keep experiments as applicable as possible to real-world scenarios. This exploration thus showcases the significance of collaborative learning techniques and the potential utilization of data from different medical institutions, to advance the accuracy and robustness of breast cancer classification models without compromising privacy.

The code is available at https://github.com/e-mny/mammo_classification to encourage further studies in this domain.

7.2 Future Directions

Due to time and technical constraints, we were unable to achieve desired results as attained by other papers. If given more time, we would look into improving the model to attain similar results. This can be done by fine-tuning the model, taking into account patient information with the scan, such as BI-RADS value, age, or including the view and laterality of the scan when training the model.

Collaboration with healthcare institutions to leverage on their data would also help the model be more robust. We noticed a difference in the types of mammograms when analysing the publicly available datasets, and this could have caused difficulty in the model being robust and learning from the visually-different mammograms. Furthermore, the lack of ground truths (other than from CBIS-DDSM), prevented the learning of what makes a mammogram malignant.

We also recognised that we undersampled two of the datasets by a significant amount. Future studies can look into this handling of imbalanced datasets in an FL environment and its impact on performance.

As mentioned earlier, we saw potential in using a model that was fine-tuned on ROI images to see comparable performance to a model with ImageNet weights, but with lower loss. This can be due to the model being more adapted to medical images instead of mere common object classes in ImageNet. The specificity of the task of mammograms is thus understood by the model, giving rise to better performances. Again, if given more ground truths and ROIs, the fine-tuning of the model could be made better with the usage of a larger dataset, and this can lead to better FL results.

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